

Customer Shopping Behavior Analysis

3,900 purchases across product categories — uncovering spending patterns, segments, and subscription behavior.



DATASET OVERVIEW

Inside the Data

3,900

Transactions

18

Columns

\$59.76

Avg Purchase

3.75

Avg Rating

Key Features

- Customer demographics: Age, Gender, Location
- Purchase details: Category, Amount, Season, Size
- Shopping behavior: Discounts, Reviews, Shipping
- 37 missing values in Review Rating

METHODOLOGY

Python EDA Pipeline



Load Data



Clean & Impute



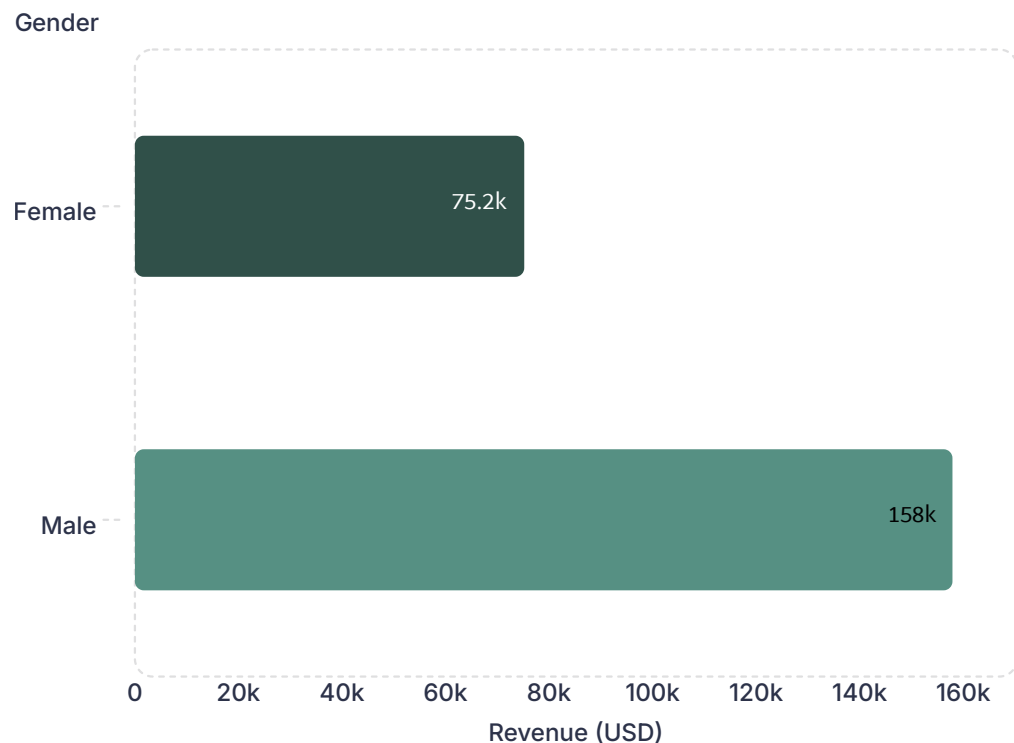
Feature Engineer



Load to PostgreSQL

Imported with pandas, imputed missing Review Ratings by category median, standardized columns to snake case, engineered age groups, and dropped redundant promo_code_used.

Revenue by Gender



Key Finding

Male customers generate **2.1x** more revenue than female customers.

839 high-spending discount users identified — customers who applied discounts yet spent above the average purchase amount.

Top-Rated & Discount-Dependent Products

Top 5 by Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rates

Product	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation

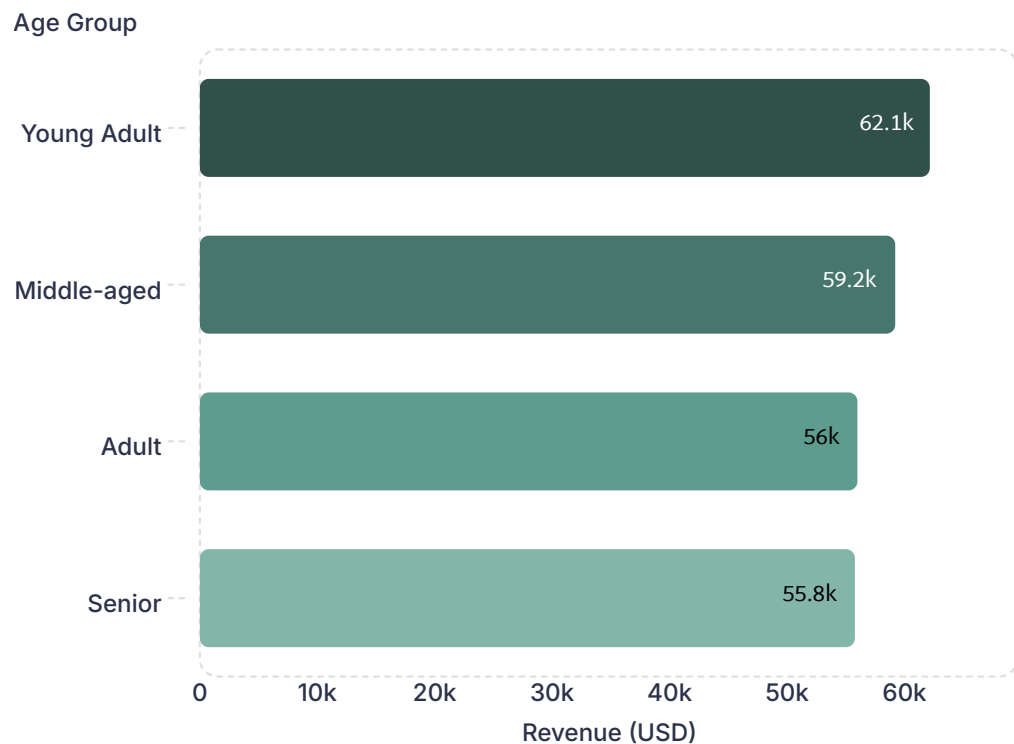


Segment Breakdown

- **Loyal:** 3,116 customers
- **Returning:** 701 customers
- **New:** 83 customers

Repeat buyers with >5 purchases: 2,518 non-subscribers vs. 958 subscribers.

Revenue by Age Group



Shipping Comparison

- **Standard:** \$58.46 avg
- **Express:** \$60.48 avg

Express shipping users spend **\$2 more** per purchase on average.

SQL INSIGHTS

Top Products per Category

Accessories

Jewelry · Sunglasses ·
Belt

Clothing

Blouse · Pants · Shirt

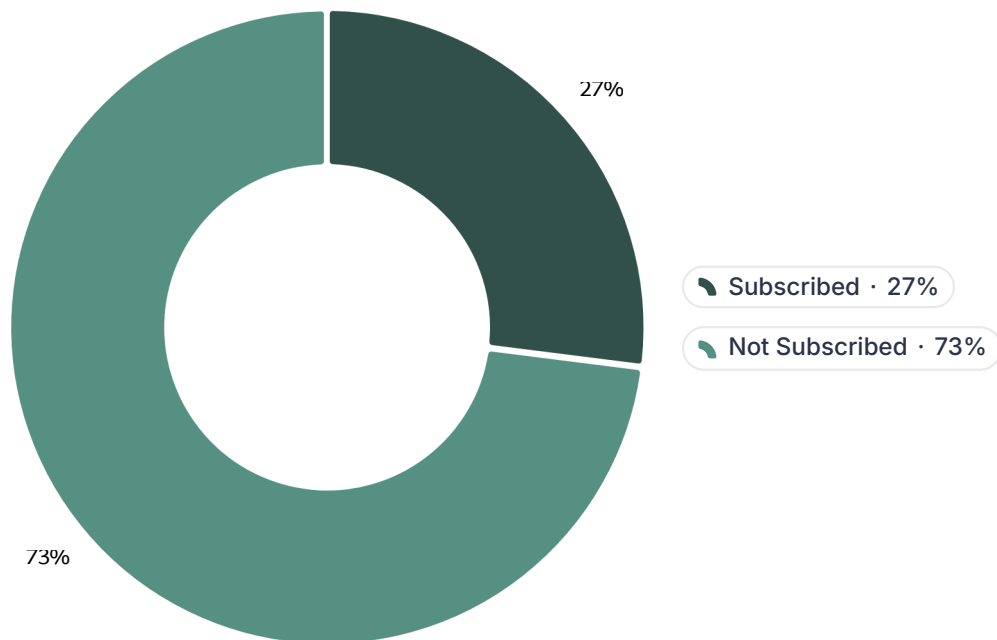
Footwear

Sandals · Shoes ·
Sneakers

Outerwear

Jacket · Coat

Subscription Status



Subscriber Spend

- **Subscribers:** 1,053 customers, \$59.49 avg spend
- **Non-subscribers:** 2,847 customers, \$59.87 avg spend

Subscription drives loyalty, not higher per-purchase spend.

RECOMMENDATIONS

Strategic Actions

Boost Subscriptions

Promote exclusive subscriber benefits.

Loyalty Programs

Reward repeat buyers to grow Loyal segment.

Review Discounts

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products.

Targeted Marketing

Focus on high-revenue age groups and express users.